



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2010

YEAR 11
Half Yearly Examination

Mathematics

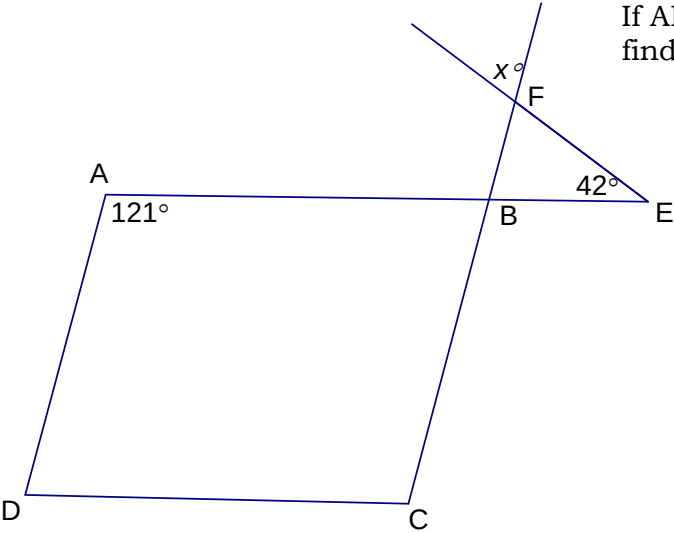
General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise instructed

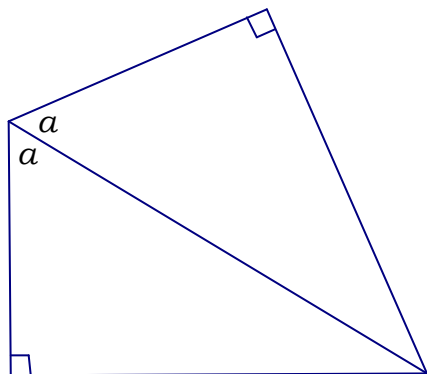
Total Marks – 100

- Attempt all questions
- All questions are NOT of equal value

Examiner: *R. Boros*

Number	Question	Marks
1.	Solve $\frac{1}{3}(x-2) - \frac{1}{2}(2-x) = 1$	2
2.	Find the value of x if $\sqrt{x} = \sqrt{50} - \sqrt{18}$	2
3.	Solve the following quadratic equation leaving your answer in surd form $(2x-1)^2 = 6$	3
4.	Express $\frac{1}{\sqrt{3}-2}$ with a rational denominator	2
5.	Expand and simplify $\sqrt{(a-5)(a+5)+25}$	2
6.	The 3 legs of a triangular sailing course for the London Olympics have lengths 8km, 10km and 16km. a) Draw a sketch showing this information. b) Mark in angle α where the smallest angle should be. c) Calculate this angle α correct to the nearest minute.	1 1 2
7.	Express $1.0\dot{2}\dot{6}$ as a rational number	2
8.	 If AD is parallel to CF, find x giving reasons.	2
9.	Find the exact value of $\tan 60^\circ \times \sin(30^\circ)$	2
10.	Evaluate $\frac{5.3}{9.6-3.7}$ correct to 2 significant figures	2
11.	What is 0.0000309 written in scientific notation?	1
12.	How many zeros are significant in the number 0.0050309?	1

13. Which congruency test would be used to establish the congruency of these 2 triangles? 1



14. Which elements in the set are rational numbers? 2

$$\left\{ \sin 30^\circ, \pi, \sqrt{10}, 3.\dot{4}, 2^{\frac{1}{2}} \right\}$$

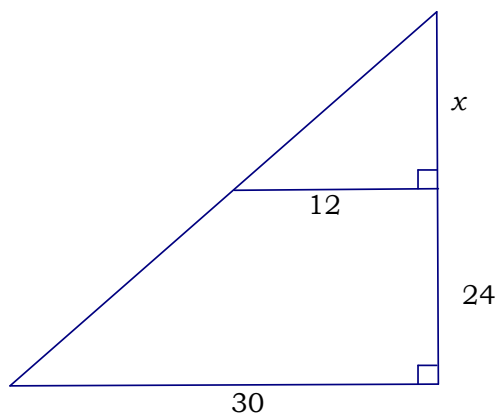
15. Which of the following statements about the diagonals of a rhombus is False? 2

- a) The diagonals bisect each other.
- b) The diagonals bisect the angles of the rhombus.
- c) The diagonals bisect at right angles.
- d) The diagonals are equal.

16. Fully factorise:

- a) $3a^2 - 13a + 12$ 2
- b) $64 - a^3$ 2
- c) $ay - ax - cx + cy$ 2

17. Find x , giving reasons/working 2



18. Solve these equations for x :

a) $\frac{3x+4}{x} = 2$ 2

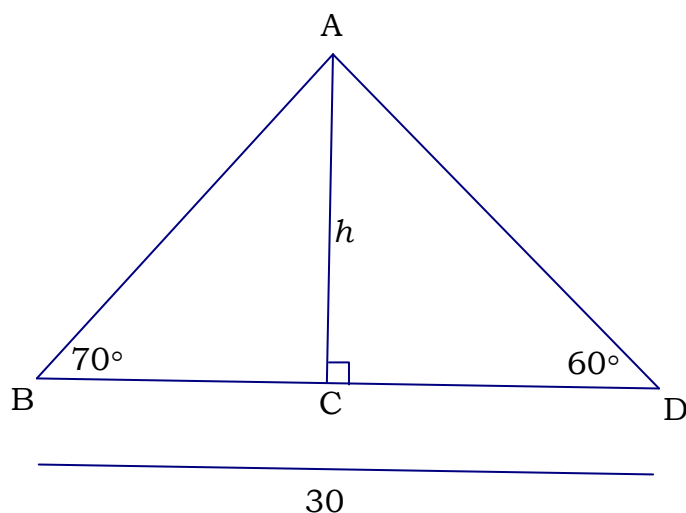
b) $x^2 = 6x$ 2

c) $3 - 2x \geq 7$ 2

19. If $f(x) = |2x - 5|$, solve $f(x) = f(4)$ 2

20. Solve for x , $|3 + 2x| \geq |x - 1|$ 3

21. Find h , correct to 2 decimal places. 3



22. Solve these simultaneous equations: 3

$$5(2x - y) = 7x + 1$$

$$3(3x + y) = 5(x - y + 12)$$

23. Each interior angle of a regular polygon is 140° . How many sides does the polygon have? 2

24. Simplify $\frac{5}{x-3} \div \frac{x^2+3x}{x^2-9}$ 3

25. Simplify $\frac{\cos A}{\sin(90^\circ - A)}$ 2

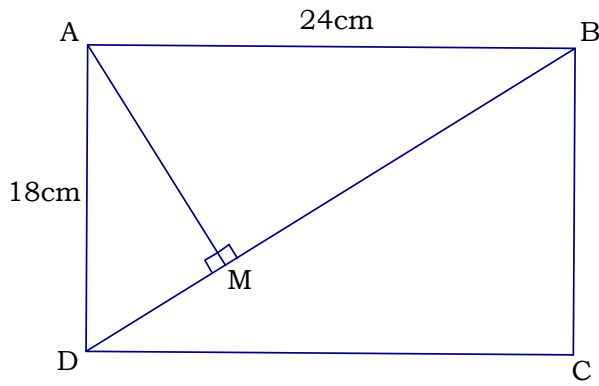
26. Given that $A = \left[\frac{9}{5}\right]^3$ 4

$$B = \left[\frac{1}{25}\right]$$

$$C = 81$$

Find the value of x and y if $\frac{A^2}{B^5C^3} = 3^x \times 5^y$

27.



ABCD is a rectangle with dimensions as shown.

- a) Find the length of BD
- b) Find the length of BM

2
2

28. A teacher is employed in 1980 at a initial salary of \$27750 p.a. After each year of service she receives an increment of \$1050 until she reaches the maximum salary of \$37200.

- a) What is her salary after 8 years of service?
- b) How long does she have to work until she receives the maximum salary?
- c) What are her total earnings for the first 10 years of service?

2
2
2

29.

By considering $0.\dot{2}\dot{9}$ as a recurring decimal which is a sum on an infinite geometric series, find the equivalent fraction to $0.\dot{2}\dot{9}$. Show all working.

2

30. Calculate the interest earned on an investment of \$11750 at 9%p.a. compounded quarterly for 5 years.

2

31. An employee invests \$950 at the beginning of each year in a superannuation scheme. Assuming interest is paid at $7\frac{1}{2}\%$ p.a. on the investment, how much to the nearest \$ will this investment grow to after 40 years?

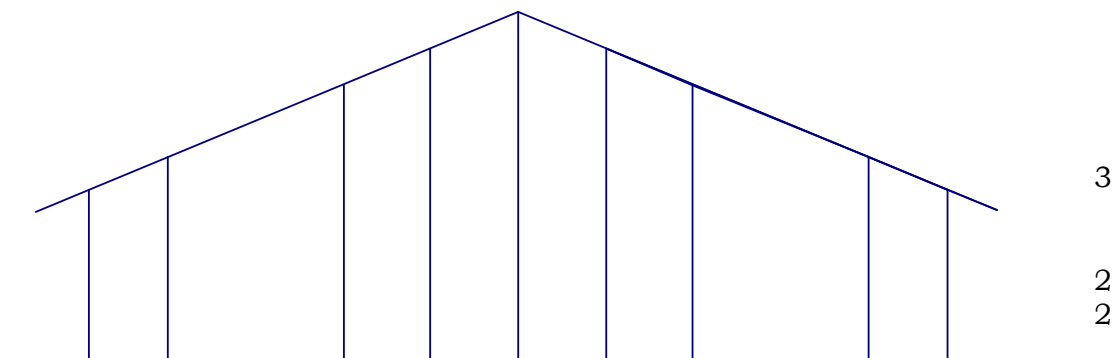
4

32. Loukia borrowed \$60000 at 18%p.a. where the interest is compounded monthly on the balance owing. If she pays off this loan in equal monthly instalments over 25 years, calculate (to the nearest cent):

- a) the amount of each monthly repayment.
- b) the total amount paid for the loan.
- c) the total interest paid
- d) the rate of simple interest (to 2 d.p.) equivalent to this compound interest.

3
2
2
2

33.



A symmetrical roof is to be supported at regular intervals by vertical posts.

The shortest posts are ' a ' metres long and consecutive posts differ in length by ' d ' metres. The total length of all posts is ' S ' metres.

Let the number of posts be $(2n+1)$.

- a) Prove that $S=dn^2+2an+a$
- b) If $S=64.4$, $d=0.1$, $a = 2$
find:
 - i) The number of posts
 - ii) The length of the longest post.

END OF EXAMINATION

Half Yearly Maths 2nd Continuers 2010

$$\textcircled{1} \quad \frac{x}{3} - \frac{2}{3} - 1 + \frac{x}{2} = 1$$

$$(\times 6) \quad \frac{6x}{3} - \frac{6 \times 2}{3} - 6 \times 1 + \frac{6 \times x}{2} = 1 \times 6$$

$$2x - 4 - 6 + 3x = 6$$

$$5x = 16$$

$$x = \frac{16}{5} = 5\frac{1}{5}, \quad 5.3 \quad \textcircled{2}$$

$$\textcircled{2} \quad \sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$$

$$\sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$$

$$\sqrt{x} = 2\sqrt{2}$$

$$= \sqrt{4 \times 2}$$

$$= \sqrt{8}$$

$$x = 8 \quad \textcircled{2}$$

$$\textcircled{3} \quad (2x-1) = \pm \sqrt{6}$$

$$2x = 1 \pm \sqrt{6}$$

$$x = \frac{1 \pm \sqrt{6}}{2} \quad \textcircled{3}$$

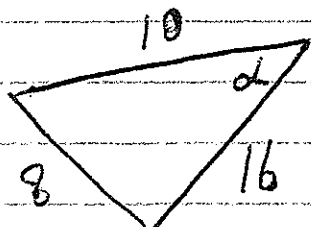
$$\textcircled{4} \quad \frac{1}{(\sqrt{3}-2)} \times \frac{(\sqrt{3}+2)}{(\sqrt{3}+2)} = \frac{\sqrt{3}+2}{3-4} = -\frac{(\sqrt{3}+2)}{1} \quad \textcircled{2}$$

$$\textcircled{5} \quad \sqrt{a^2 - 25 + 25} = \sqrt{a^2} = a \quad \textcircled{2}$$

⑥

(a)

(b)



2

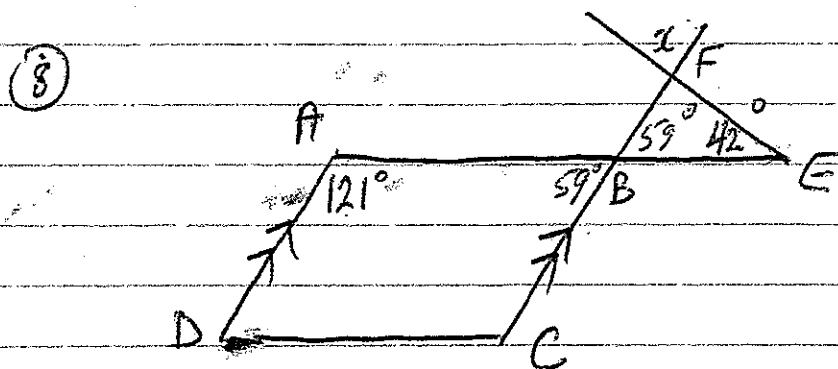
6 (c) cosine rule. $8^2 = 10^2 + 16^2 - 2 \times 10 \times 16 \times \cos \alpha$
 $64 = 356 - 320 \cos \alpha$
 $-292 = -320 \cos \alpha$
 $\cos \alpha = 0.9125,$
 $\alpha = 24^\circ 9' \quad (2)$

7 $1.\dot{0}2\dot{6} = 1.0262626 \dots$
 let $x = 0.02626 \dots$
 $100x = 2.62626$

$99x = 2.6$

$x = \frac{2.6}{99} = \frac{26}{990} = \frac{13}{495}$

so $1.\dot{0}2\dot{6} = 1 \frac{13}{495} \quad (2)$



$\hat{ABC} = 180 - 121 = 59^\circ$ co-interior angles.

$\hat{FBE} = 59^\circ$ vertically opposite

$\hat{BFE} = x = \text{vertically opposite} = 180 - (59^\circ + 42^\circ)$
 $= 79^\circ \quad (2)$

9 $\sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2} \quad (2)$

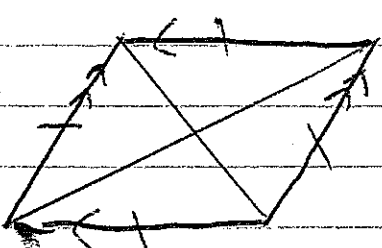
10 0.90 25F (2)

11 $3.09 \times 10^{-5} \quad (1)$

12 2 (1)

(13) SSS SAS AAS RHS (1)
X X ✓ X

(14) $\{\sin 30^\circ, 3.4\}$ (2)

(15)  a) T
b) T
c) T
d) F (2)

(16) a) $(3a-4)(a-3)$ (2)
b) $(4-a)(16+4a+a^2)$ (2)
c) $\frac{a(y-x) + c(y-x)}{(y-x)(a+c)}$ (2)

(17) small and larger Δ are similar, 2 angle test.
ratio of sides -

$$\frac{x}{24+x} = \frac{12}{30}$$

$$30x = 288 + 12x$$

$$18x = 288$$

$$x = 16 \quad (2)$$

(18) a) $3x+4=2x$
 $x=-4$ (2)

(c) $3-2x \geq 7$
 $-2x \geq 4$

(b) $x^2 = 6x$
 $x^2 - 6x = 0$
 $x(x-6) = 0$
 $x=0, x=6$ (2)

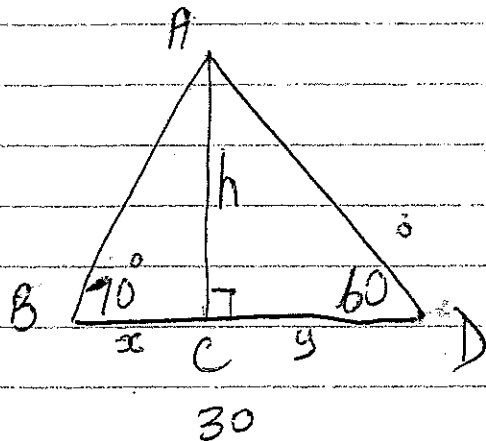
$x \leq -2$ (2)

$$(19) f(x) = |2x - 4 - 5| = 3 \quad (2)$$

$$(20) \begin{aligned} 3 + 2x &\geq x - 1 & \text{and} & \quad 3 + 2x \leq -(x - 1) \\ x &\geq -4 & & \quad 3 + 2x \leq -x + 1 \\ & & & \quad 3x \leq -2 \\ & & & \quad x \leq -\frac{2}{3} \end{aligned}$$

$$-4 \leq x \leq -\frac{2}{3} \quad (3)$$

(21)



Since $x + y = 30$

$$\frac{h}{\tan 70^\circ} + \frac{h}{\tan 60^\circ} = 30$$

$$0.36397h + 0.57735h = 30$$

~~$$0.94132h = 30$$~~

~~$$h = \frac{30}{0.94132}$$~~

$$h = \frac{30}{0.94132} \quad (3)$$

$$\tan 70^\circ = \frac{h}{x}$$

$$x \tan 70^\circ = h$$

$$x = \frac{h}{\tan 70^\circ}$$

$$\tan 60^\circ = \frac{h}{y}$$

$$y \tan 60^\circ = h$$

$$y = \frac{h}{\tan 60^\circ}$$

$$(22) \quad 10x - 5y = 7x + 1 \Rightarrow 3x - 5y = 1 \quad (1)$$

$$9x + 3y = 5x - 5y + 60 \Rightarrow 4x + 8y = 60 \quad (2)$$

$$(1) \times 4 \quad 12x - 20y = 4 \quad -$$

$$(2) \times 3 \quad 12x + 24y = 180$$

$$-44y = -176$$

$$y = 4$$

(3)

$$\text{and } 3x - 20 = 1 \quad x = 7$$

(23) Interior is 140° exterior is $180^\circ - 140^\circ = 40^\circ$
 total exterior $\frac{360^\circ}{40^\circ} = 9$ sides. (2)

(24)
$$\frac{5}{(x-3)} \times \frac{(x-3)(x+3)}{x(x+3)} = \frac{5}{x} \quad (3)$$

(25)
$$\frac{\cos A}{\cos A} = 1 \quad (2)$$

(26)
$$\frac{9^6}{5^6} = \frac{3^{12}}{5^6} \times \frac{5^{10}}{3^{12}} = 3^0 \times 5^4$$

$$\frac{1}{25^5} \times 81^3$$

 $x=0, y=4 \quad (4)$

(27) (BD) = $18^2 + 24^2$
 (a) BD = 30 (2)

(b) Area $\triangle ABD = \frac{1}{2} \times 24 \times 18 = 216$

so $216 = \frac{1}{2} \times 30 \times AM$
 $AM = 14.4$

thus $(14.4)^2 + (BM)^2 = 24^2$
 $BM = 19.2 \quad (2)$

(28) (a) $a = 27750$

$d = 1050$

$n = 8$

$U_n = a + (n-1)d$
 $= 27750 + 7 \times 1050$
 $= \$35100 \quad (2)$

(b) $37200 = 27750 + (n-1)1050$

$9450 = 1050(n-1)$

$9 = n-1$

$n = 10$ years. (2)

(c) $S_n = \frac{n}{2}(a+l)$
 $= \frac{10}{2}(27750 + 37200)$
 $= \$324750 \quad (2)$

$$(29) \quad 0.29 = 0.29 + 0.0029 + 0.000029 \dots$$

$$r = \frac{1}{100}, \quad a = 0.29$$

$$S_{\infty} = \frac{a}{1-r} = \frac{0.29}{1-\frac{1}{100}} = \frac{0.29}{.99} = \frac{29}{99} \quad (2)$$

$$(30) \quad A = P\left(1 + \frac{r}{100}\right)^n$$

$$= 11750\left(1 + \frac{2.25}{100}\right)^{20} = \$18335.98$$

$$\text{interest is } \$6585.98 \quad (2)$$

$$(31) \quad \text{Year 1 } A = 950\left(1 + \frac{7.5}{100}\right)^{40}$$

$$\text{Year 2 } A = 950\left(1 + \frac{7.5}{100}\right)^{39}$$

$$\text{Year 40 } A = 950\left(1 + \frac{7.5}{100}\right)^1$$

$$\text{total} = 950(1.075)^1 + 950(1.075)^2 + \dots + 950(1.075)^{40}$$

$$= 950(1.075)^1 [1 + 1.075^1 + \dots + 1.075^{39}]$$

$$\text{sum } [] \quad S_n = \frac{rL - a}{r-1} = \frac{1.075 \times 1.075^{39} - 1}{1.075 - 1}$$

$$\text{total is } \$232086 \text{ (nearest \$)} = \frac{1.075^{40} - 1}{.075} \quad (4)$$

$$(32) \quad \$60000, \quad 18\% \text{ pa} = 1.5\% \text{ p month} \quad 300 \text{ months}$$

$$\text{After 1 month: } A_1 = 60000\left(1 + \frac{1.5}{100}\right) - m$$

$$= 60000(1.015) - m$$

(a)

$$A_2 = 60000(1.015)^2 - 1.015m - m$$

$$= 60000(1.015)^2 - m(1 + 1.015)$$

$$A_{300} = 60000(1.015)^{300} - m(1 + 1.015 + \dots + 1.015^{299})$$

$$m = \frac{60000(1.015)^{300}}{1 + 1.015 + \dots + 1.015^{299}}$$

$$\text{bottom } S_n = \frac{rL - a}{r - 1} = \frac{1.015 \times 1.015^{299} - 1}{1.015 - 1} \\ = \frac{1.015^{300} - 1}{0.015} = 5737.25$$

$$\text{total } m = \$910.46 \quad (3)$$

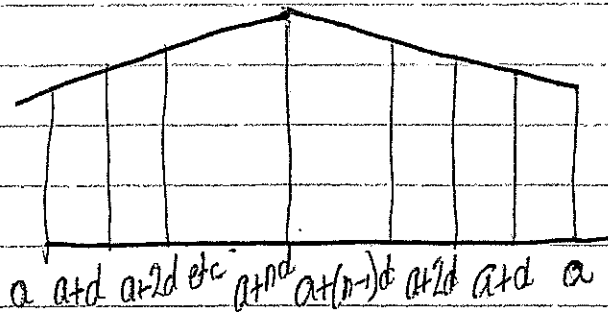
$$(b) \$910.46 \times 300 = \$273138 \quad (2)$$

$$(c) \$273138 - \$60000 = \$213138 \quad (5)$$

$$213138 = 60000 \times \frac{r}{100} \times 25$$

$$r = 14.21\% \text{ p.a. flat} \quad (2)$$

(33) (a)



$$S_n = \frac{n}{2}(2a + (n-1)d) \times 2 + a + nd \\ = n(2a + nd - d) + a + nd \\ = 2an + n^2d - nd + a + nd \\ = dn^2 + 2an + a \quad (3)$$

$$(b) \quad 64 \cdot 4 = 0.1n^2 + 2 \times 2n + 2$$

$$62.4 = 0.1n^2 + 4n$$

$$\times 10 \quad 624 = n^2 + 40n$$

$$n^2 + 40n - 624 = 0$$

$$\text{quad formula} \quad n = \frac{-40 \pm \sqrt{1600 - 4 \times 1 \times -624}}{2}$$

$$= \frac{-40 \pm 64}{2}$$

$$\text{take (+)} \quad \frac{-40 + 64}{2} = 12 \quad (2)$$

$$(c) \text{ longest post a+nd} \quad = 2 + 12 \times 0.1$$
$$= 3.2 \text{ m.} \quad (2)$$